

Transformers



Attention Mechanism

Attention in Human Visual Processing System

552 HARRY POTTER

the window. Dumbledore watched her fly away, and as her silvery glow faded he turned back to Snape, and his eyes were full of tears.

'After all this time?'

'Always,' said Snape.

And the scene shifted. Now, Harry saw Snape talking to the portrait of Dumbledore behind his desk.

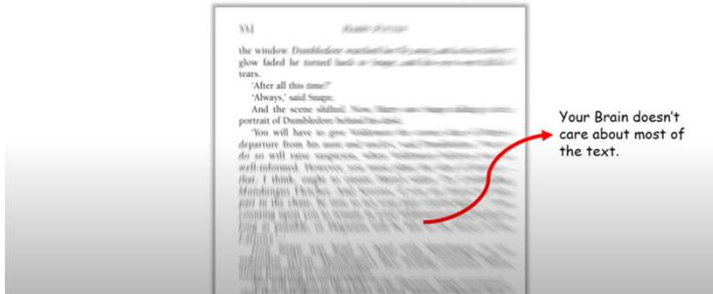
'You will have to give Voldemort the correct date of Harry's departure from his aunt and uncle's,' said Dumbledore. 'Not to do so will raise suspicion, when Voldemort believes you so well-informed. However, you must plant the idea of decoys – that, I think, ought to ensure Harry's safety. Try Confusing Mundungus Fletcher. And Severus, if you are forced to take part in the chase, be sure to act your part convincingly ... I am counting upon you to remain in Lord Voldemort's good books as long as possible, or Hogwarts will be left to the mercy of the Carrows ...'

Now Snape was head to head with Mundungus in an unfamiliar tavern, Mundungus's face looking curiously blank, Snape frowning in concentration.

'You will suggest to the Order of the Phoenix,' Snape murmured, 'that they use decoys. Polyjuice Potion. Identical Potters. It is

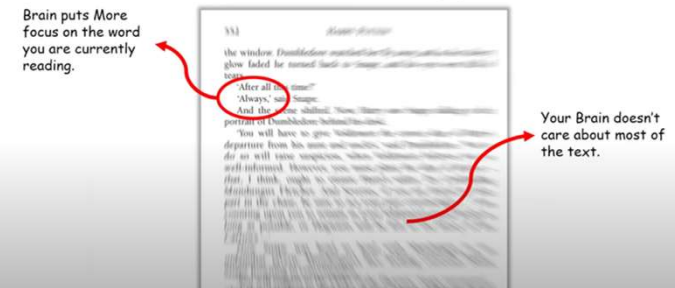
Attention Mechanism

Attention in Human Visual Processing System



Attention Mechanism

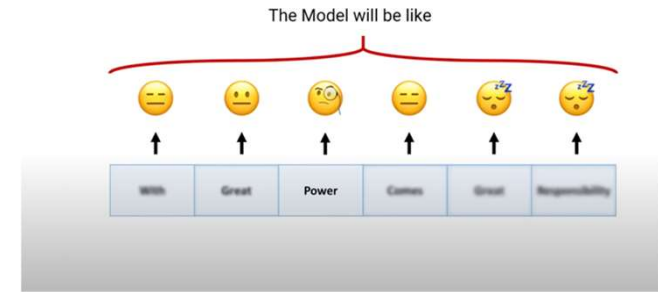
Attention in Human Visual Processing System



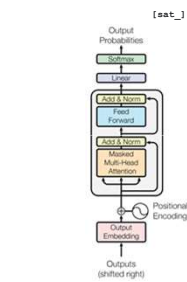
Attention Mechanism



Attention Mechanism

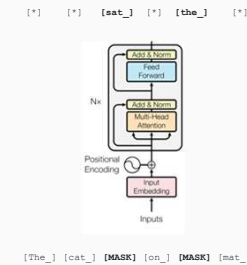


Decoder-only GPT

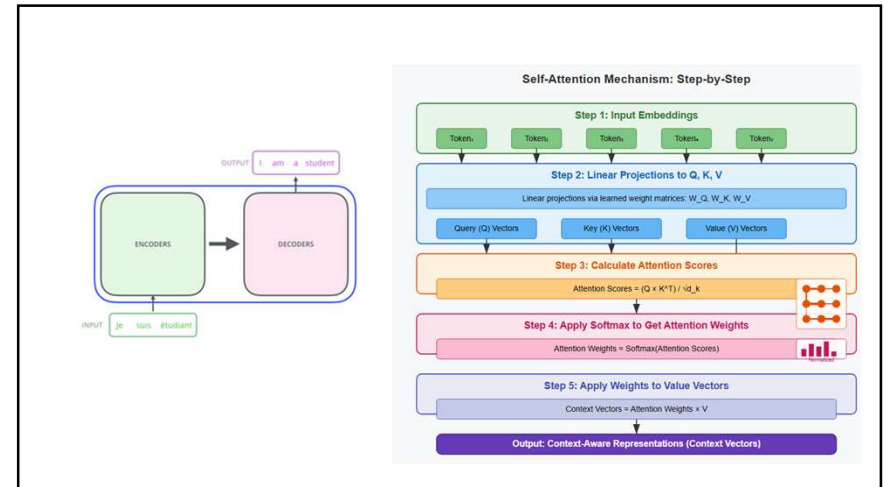
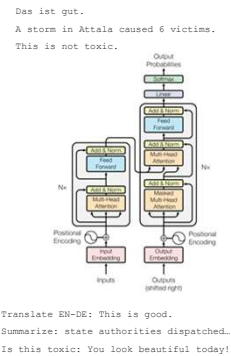


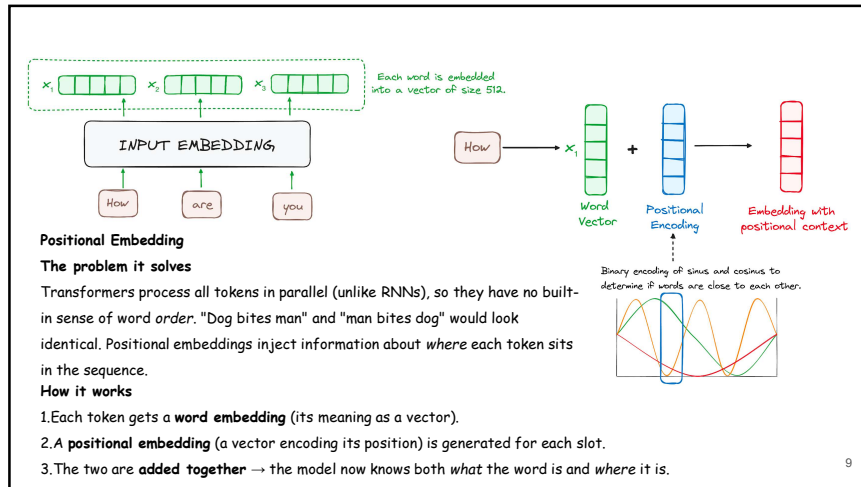
Transformer source: "Attention is All You Need" paper

Encoder-only BERT

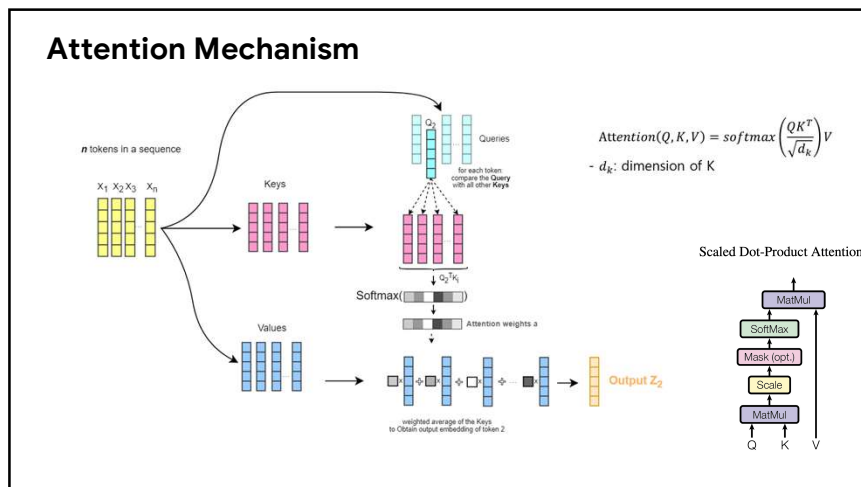
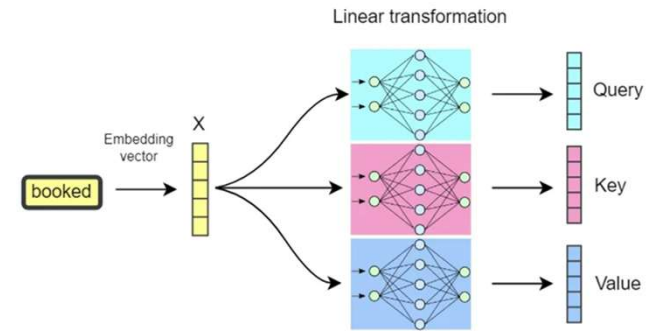


Enc-Dec T5

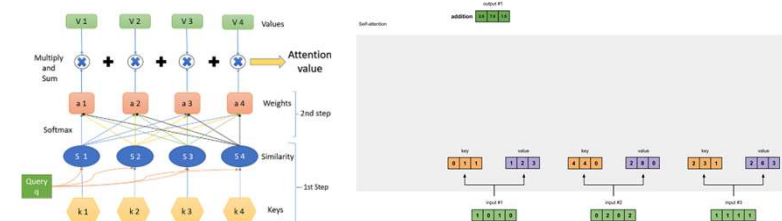




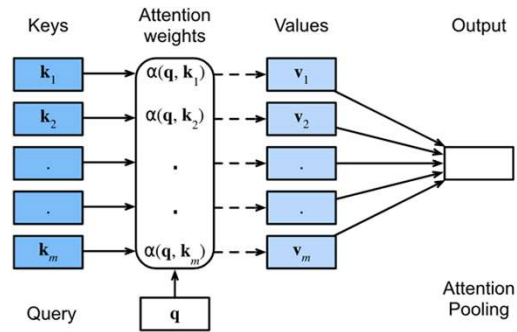
Attention Mechanism



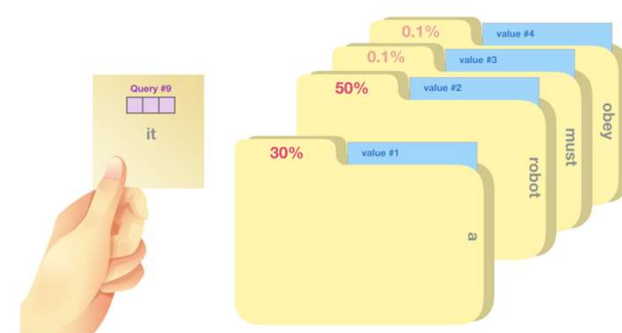
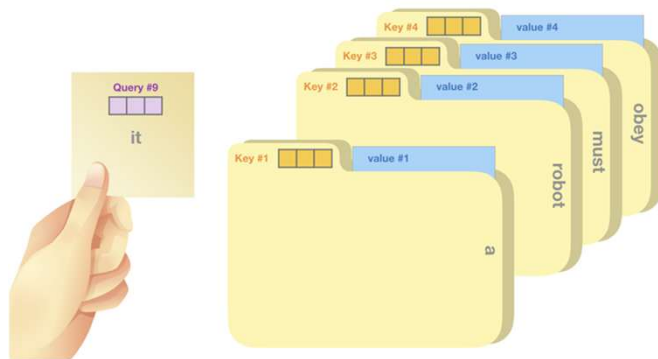
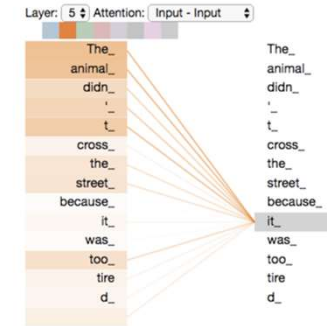
SELF ATTENTION



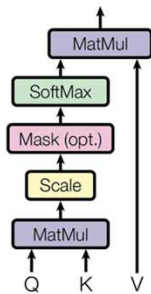
Attention Mechanism



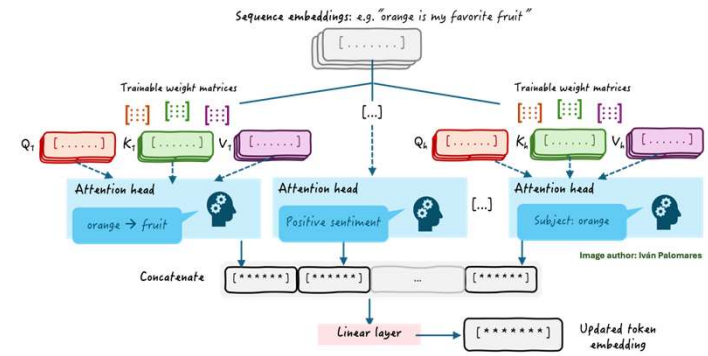
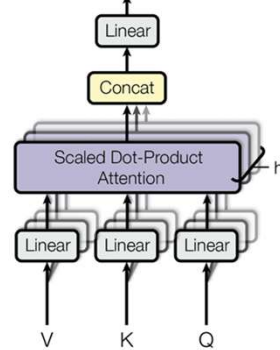
Attention!!



Scaled Dot-Product Attention

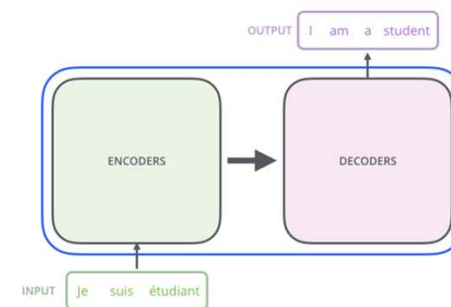


Multi-Head Attention

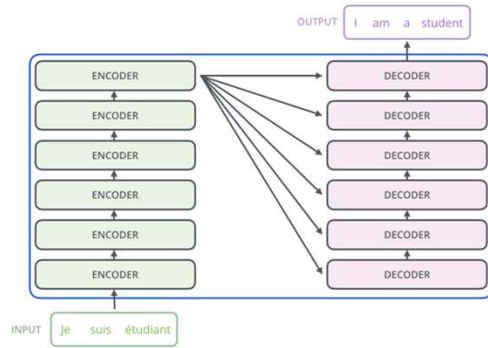


Word	Value vector	Score	Value X Score
<S>		0.001	
a		0.3	
robot		0.5	
must		0.002	
obey		0.001	
the		0.0003	
orders		0.005	
given		0.002	
it		0.19	
		Sum:	

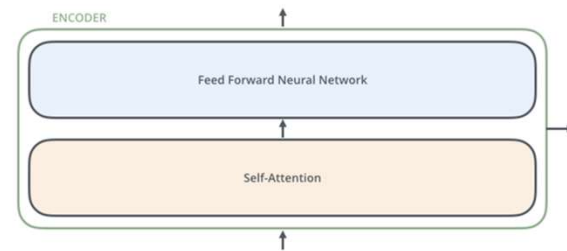
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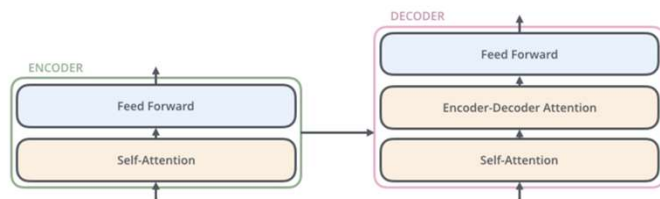
Transformers



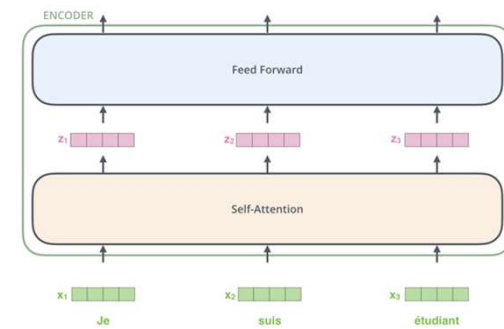
Transformers



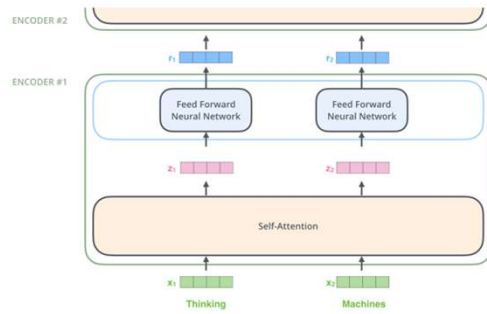
Transformers



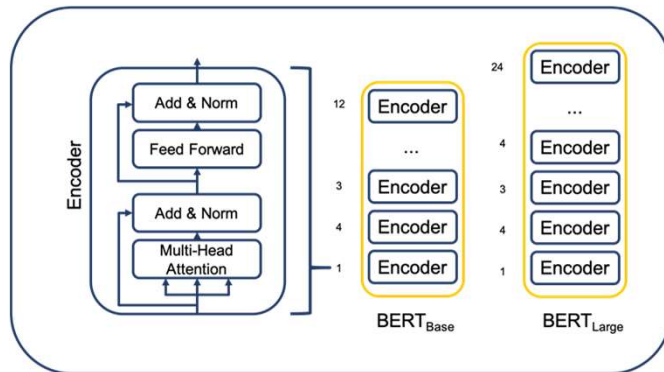
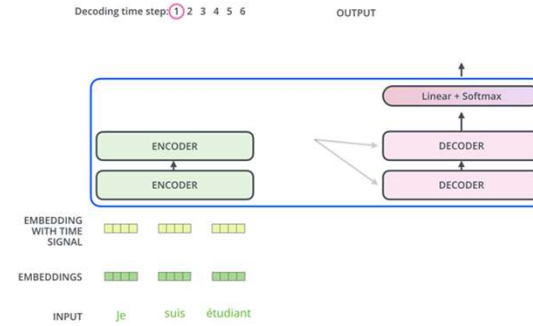
Transformers



Transformers

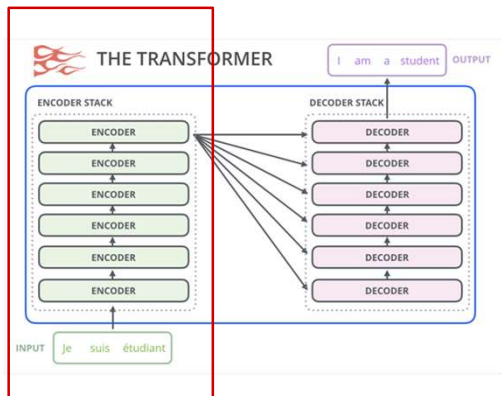


Transformers



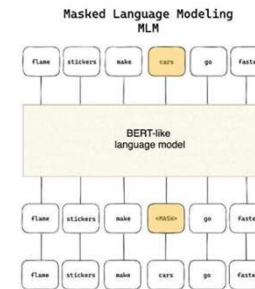
Bert Model

The BERT model is trained on two tasks: Masked Language Modeling, where the objective is to predict the masked word based on the context

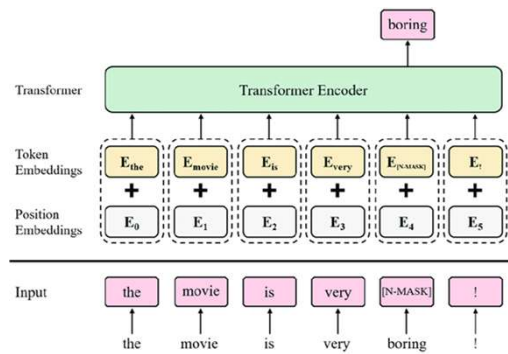


Pre-Training

Understanding word in the context → Understanding the context



BERT was pre-trained with two unsupervised tasks, Masked LM and Next Sentence Prediction.



BERT's encoder-only architecture, context-dependent word embeddings

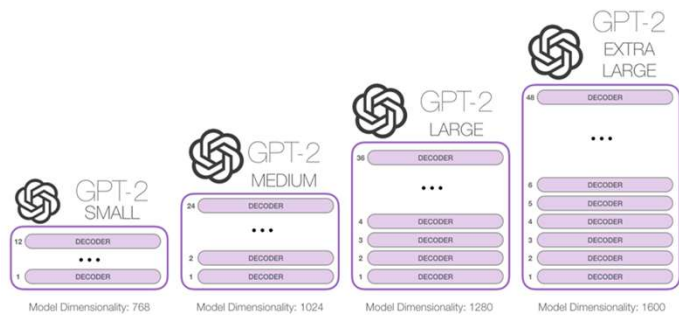
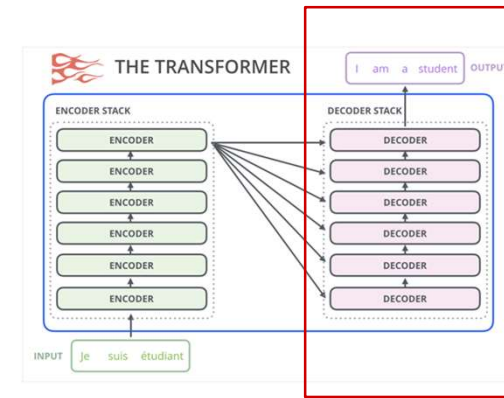
I like to this shirt



Ground truth
Embedding layer
Attention
Predictions

GPT Models

	PROPRIETARY					
						
	GPT-1 (130M)	GPT-2 (1.5B)	GPT-3 (175B)	LaMDA (137B)	MT-NLG (530B) PaLM (540B)	GPT-4 (1T)
					 Chinchilla (70B) Gopher (280B)	
	Megatron-LM (8.3B)	Turing-NLG (70B)	Jurassic-1 (178B)			
	OPEN SOURCE					
					 Godel (175B)	
	ELMo (84M)	BERT (340M)	T-ELMo (485M)	GPT-J (6B)	BLOOM (70B)	LLaMa (65B)
	Ai2		Ai2			
	2018	2019	2020	2021	2022	2023



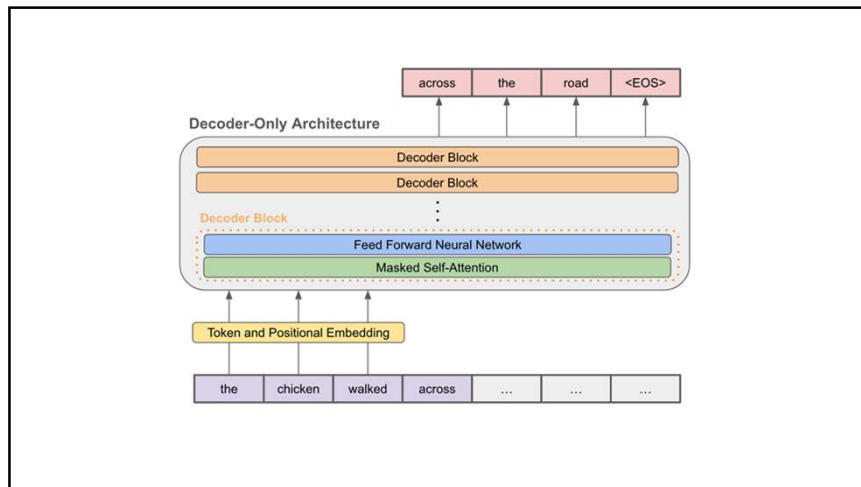
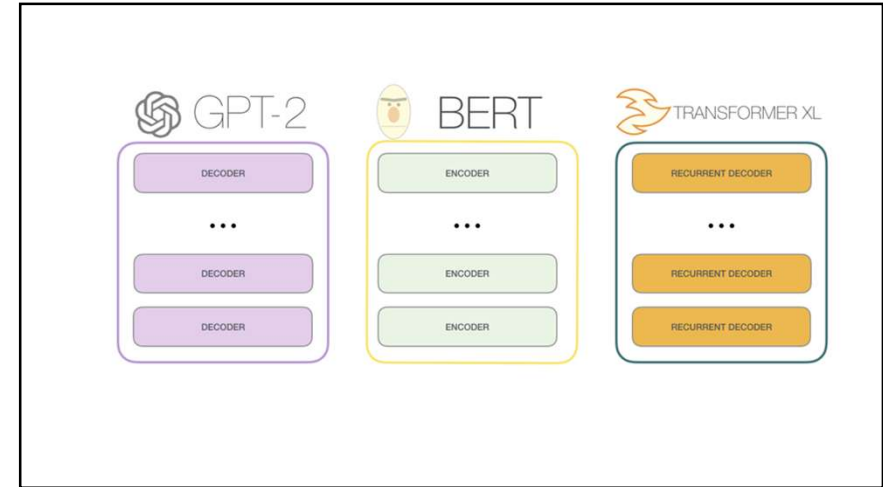
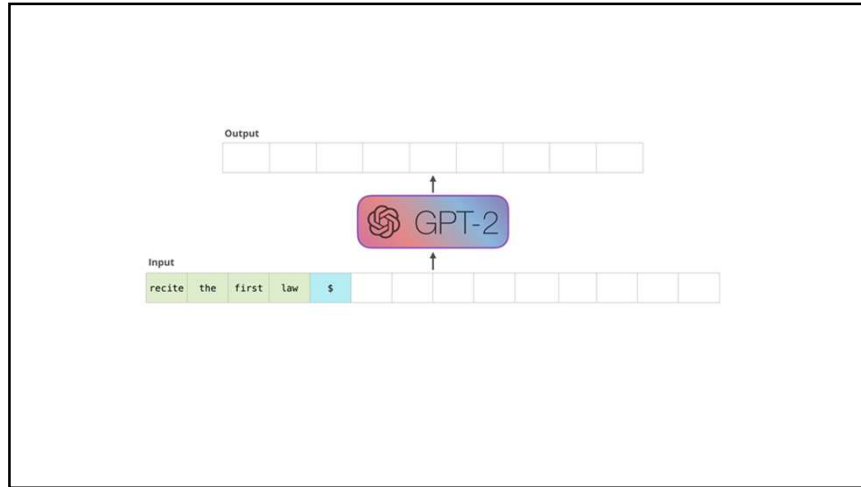
GPT is based on the decoder-only Transformer Architecture

Pre-Training

Predicting the next word -> Grasping the flow of dialogue



GPT was pre-trained using Autoregressive Language Modeling for contextual prediction



Chat-GPT Based Models - Instruct Tuned

